

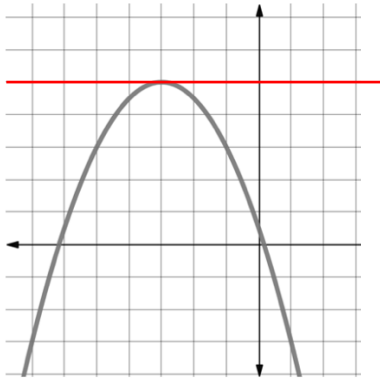
Algebra 1
Unit 9
Lesson 5 Homework

Name Answer key
Date _____
Period _____

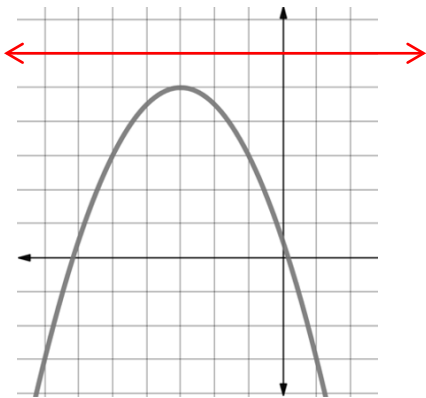
Draw a line on the graph of the quadratic where there would be...

1. ...one real solution.

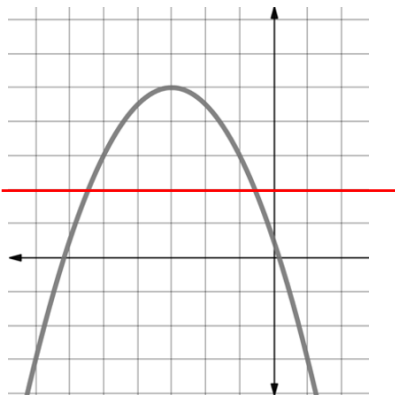
Answers may vary



2. ...no real solution.



3. ...two real solutions.



Solve each equation. Show your work.

$$4. \quad \frac{6x^2}{6} = \frac{150}{6} \qquad \sqrt{x^2} = \sqrt{25}$$

$$x^2 = 25 \qquad x = \pm 5$$

$$5. \quad \frac{-32}{-2} = \frac{-2x^2}{-2}$$

$$\sqrt{16} = \sqrt{x^2} \\ \pm 4 = x$$

$$6. \quad \frac{8}{-3} = \frac{x^2 + 3}{-3}$$

$$\sqrt{5} = \sqrt{x^2} \\ x = \pm \sqrt{5}$$

$$7. \quad \frac{2(x^2 - 7)}{2} = \frac{-6}{2}$$

$$\sqrt{x^2} = \sqrt{4}$$

$$\frac{x^2 - 7}{+7} = \frac{-3}{+7} \qquad x = \pm 2$$

$$8. \quad 4 \cdot \frac{1}{4}x^2 = 36 \cdot 4$$

$$\sqrt{x^2} = \sqrt{144} \\ x = \pm 12$$

$$9. \quad \frac{2x^2 + 5}{-5} = \frac{65}{-5}$$

$$\sqrt{x^2} = \sqrt{30}$$

$$\frac{2x^2}{2} = \frac{60}{2} \qquad x = \pm \sqrt{30}$$

$$10. \quad \frac{x^2 - 25}{+25} = \frac{24}{+25}$$

$$\sqrt{x^2} = \sqrt{49} \\ x = \pm 7$$